

Improving Medical Visual Instruction Tuning with Labeled Datasets

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Abstract. We introduce the Medical Vision Instruction-tuning Dataset (MVID), a large-scale dataset containing 558,052 samples designed to enhance the instruction-following capabilities of medical vision-language models (VLMs). Previous vision-language datasets did not include clinical data as they were generated by commercial large language models (LLMs) that did not meet the privacy and regulatory constraints. Additionally, prior instruction-tuning efforts for medical VLMs focused on figure-caption pairs from PubMed, covering a broad array of medical topics but lacking clinical applications related themes, such as knowledge, detection, and localization of common findings. To address these limitations, MVID is generated with an open LLM and incorporates a diverse range of labeled data, including classification, segmentation, and free-text tasks, equipping models with essential medical competencies. Our experiments show that medical VLMs trained on MVID outperform those trained on PubMed-based datasets on VQA benchmarks by up to 20.29% in a zero-shot setting. Furthermore, our 7 billion parameter model demonstrates stronger performance than GPT-4o on radiology tasks, highlighting the effectiveness of MVID in developing capable medical VLMs.

Keywords: Visual Instructions · Multimodal Large Language Model · Medical Visual Question Answering.

1 Introduction

Medical Vision Language Models (VLMs) have the potential to serve as generalist assistants in clinical workflow due to their zero-shot abilities [27]. While proprietary medical VLMs seem promising [34], the effectiveness of open-source alternatives remains questionable [16]. However, specifically in the clinical domain, open-source models are usually preferred due to privacy regulations. Open-source approaches often rely on the reformulation of PubMed figure captions to visual instructions using GPT models [24]. This poses several issues for clinical use-cases: (1) PubMed articles include a broad range of biomedical topics, which do not necessarily align with common clinical situations. (2) Precision is crucial for clinical applications, but the automatic linking of PubMed figures and captions is noisy [6] (3) GPT cannot be applied to private data.

A practical solution for quality issues is creating visual instructions leveraging existing medical imaging datasets to which a plethora of expert-created classification labels, segmentation masks, bounding boxes, and imaging reports are attached. Additionally, advances in open-weight LLMs have reduced the gap between proprietary models [8], offering an alternative to GPT-based visual instruction generation.

In this paper we utilize large collections of medical imaging data annotated by experts and specialized models. We introduce a methodology that repurposes these high-quality annotations to generate reliable visual instructions for medical VLM training. Additionally, we utilize a locally deployed LLM, ensuring that our method remains applicable to private clinical datasets while maintaining high standards of data security and compliance.

Our contributions are threefold: (1) We generate and publicly release MVID¹, a dataset comprising **558,052 visual instruction samples** derived from labeled medical imaging datasets using an open-weight LLM; (2) We train medical VLMs on MVID and demonstrate **up to 20.29% higher average accuracy** compared to models trained on PubMed-derived instructions across multiple medical visual question answering (VQA) benchmarks; (3) We introduce and release² **new medical VLMs with 7 billion parameters that perform competitively with GPT-4o**, advancing the development of clinically applicable, privacy-compliant, and open-source medical VLMs.

2 Related Work

The LLaVA approach [26] has been shown to produce well-performing models on various benchmarks using a simplistic architecture: A two-layer perceptron that maps image embeddings to the text embedding space. LLaVA relies on visual instruction tuning data generated by GPT-4, which is derived from existing image captions. Following its success, this approach was extended to the medical domain by applying the same methodology to PubMed figure captions, leading

¹ <https://huggingface.co/datasets/ikim-uk-essen/MVID>

² <https://huggingface.co/ikim-uk-essen/MVID-Qwen2.5-7B>

to the development of Llava-Med [24]. More recently, improvements have been made by querying larger commercial VLMs to refine the generated visual instruction tuning data [35] or to create new visual instruction tuning datasets [6]. However, these enhancements can only be applied to non-private data, limiting their applicability to clinical datasets.

3 Medical Visual Instruction Dataset

We compiled MVID, a dataset of 558,052 medical visual instructions collected from 29 medical imaging datasets spanning nine medical domains and five types of annotations, as detailed in Table 1. We apply different methods to construct textual descriptions of the given annotations. Based on these descriptions, we generate conversations between physicians and a VLM using the open-weight LLM *Llama-3.3-70B-Instruct* [8]. An overview is illustrated in Figure 1. The following two sections describe the details of the generation process.

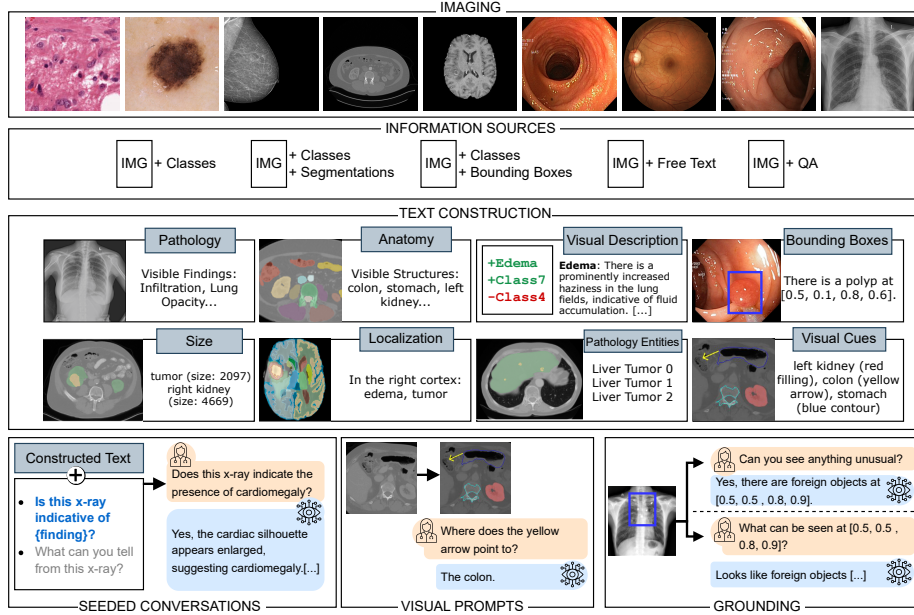


Fig. 1. Data generation process of MVID. We include data from 9 different domains with varying types of labels serving as information sources. We then derive enriched textual descriptions by applying multiple methods. These descriptions are then fed to an LLM to generate visual instructions with prompting methods.

3.1 Construction of textual descriptions

For all classification, segmentation and bounding box datasets we list the given pathological and anatomical features. Some labels correspond to specific visual properties. For example, the presence of a pleural effusion is indicated by noticeable blunting of the costophrenic angles. Where applicable, we formulated **Visual Descriptions** of such labels to enhance the available information for generating visual instructions.

For CT and MRI datasets, we sample individual slices. For the pathology dataset, we ensure an equal distribution of slices with and without pathologies. In the case of Atlas [15], a CT dataset containing segmentations of 142 anatomical structures, we sample one slice for each unique combination of segmentation masks. We derived various forms of spatial information from segmentation masks:

Mask Sizes We include the segmentation size in pixels in the textual descriptions. When generating visual instructions, we instruct the LLM to include this information without mentioning specific pixel sizes (e.g., how large a tumor is in relation to the affected organ).

Pathology Localization To enhance pathology localization, we generated anatomical segmentation masks using TotalSegmentator [38] for CT images and SynthSeg [4] for brain MRIs. Pathology segmentation masks often lacked detailed anatomical context—for example, the RibFrac dataset [18] annotated different types of rib fractures but did not specify the affected rib. By combining generated anatomical masks with pathology masks, we enriched the textual descriptions with anatomical information.

Pathology Entities We extracted individual pathology entities by computing the connected components of pathology segmentation masks. This allowed us to determine both the number of pathologies present and their respective sizes. When applicable, these entities were also mapped to anatomical regions using the methods described earlier, providing additional spatial context for visual instruction generation.

3.2 Visual Instruction Generation

Our dataset generation process includes specialized prompts for each dataset, providing contextual explanations and outlining the expected visual instructions. These prompts were refined iteratively by examining generated visual instruction samples. Additionally, we applied different prompting strategies to formulate diverse visual instructions from the constructed textual descriptions.

Conversation Seeds While using a temperature of 1.0 allows for variability in generated conversations, we found that the outputs were often identical for the same inputs, especially, in classification datasets with a limited number of labels, leading to repetitive results. To address this, we use conversation seeds—a set of introductory statements in conversations. For classification datasets, these seeds included placeholders that were randomly replaced with a possible class label (e.g., "Does this X-ray show signs of {finding}?", where finding is replaced with

a randomly selected class for each conversation). To ensure diversity, each generated conversation was forced to begin with a randomly selected conversation seed.

Visual Prompts To enhance the visual instruction tuning dataset, we introduced visual prompts by incorporating segmentation masks with distinct **Visual Cues**. These cues were assigned randomly from a predefined color set and presented in various forms, such as arrows, contours, or filled regions. The visual prompts were overlaid onto the images, with their color and shape included in the corresponding generation prompts.

Groundings To enable visual grounding in generated conversations, we incorporated datasets with **Bounding Box** annotations. Each bounding box was represented using the corresponding label followed by a unique identifier tag (e.g., Polyp<Bounding Box 1>). These tags were then included in the input information for visual instruction generation. The LLM was explicitly instructed to utilize these tags, either by integrating them into user queries or by referencing them in the VLM’s responses. As a post-processing step, the grounding tags were replaced with the corresponding bounding box coordinates $[X_{min}, Y_{min}, X_{max}, Y_{max}]$, normalized to a range between $[0, 1]$.

4 Experiments

To assess the efficiency of MVID, we train multiple VLMs and compare their performance with existing models. Our experiments are structured into two parts:

Experiment 1: Direct Comparison with Llava-Med We train a model on MVID using the same configuration as LLaVA-Med-v1.5 [24]: The LLaVA 1.5 architecture [26], a 336×336 CLIP-Large image encoder [33] and the Mistral-7B-Instruct-v0.2 LLM³. We compare this configuration to a 224×224 BioClip image encoder [40], the biomedical LLM Biomistral [22] and Qwen2.5-7B-Instruct [39]. After identifying the best configuration, we train a model on the LLaVA-Med dataset for further comparison. We select a random subset of 60,000 conversations from MVID to ensure a fair comparison to LLaVA-Med.

Experiment 2: Full MVID Training: Building on the insights from Experiment 1, we train a model on the entire MVID dataset and compare its performance against various existing VLMs.

4.1 Training

For a direct comparison with LLaVA-Med, we train all models on the LLaVA-Med alignment dataset using a learning rate of $2e^{-3}$, a cosine learning rate schedule, and a batch size of 256 for one epoch in Stage 1. In Stage 2, we set the learning rate to $5e^{-6}$. In our second experiment on the full MVID dataset, we incorporate the non-medical LLaVA-1.5 alignment dataset as well as the LLaVA-v1.5 instruction dataset. All experiments were conducted on a single NVIDIA DGX node with $8 \times H100$ GPUs.

³ <https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.2>

Table 1. All labeled imaging datasets that were used for the generation of MVID. **#Conv.** refers to the number of visual instruction conversations that were generated for each dataset. The available labels include visual question-answer pairs (VQA), classification labels (Cls), segmentation masks (Seg.) and bounding boxes (BB). The augmentations are described in Section 3.1.

Dataset	#Conv.	Modality	Labels	Augmentations
MIMIC-CXR-VQA [3]	100,000	X-Ray	VQA	-
Long Tail CXR [14]	88,336	X-Ray	Cls	Seeds, Visual Desc.
Atlas Dataset [15]	82,950	CT	Seg.	Seeds, Visual Prompts
MIMIC-CXR [19]	60,720	X-Ray	Text	Report Filtering
TCGA-UT [20]	30,255	Pathology	Cls	-
Fitzpatrick [10,11]	16,510	Dermatology	Cls	-
Dermnet ⁴	15,435	Dermatology	Cls	-
VinDr-CXR [29]	13,123	X-Ray	BB	Groundings
MSD Liver [1]	10,674	CT	Seg.	Spatial Inf.
VinDr-SpineXR [30]	10,340	X-Ray	Cls	Seeds, Visual Desc.
MSD Brain Tumor [1]	10,282	MRI	Seg.	Seeds, Spatial Inf., Anatomy
MSD HepaticVessel [1]	10,244	CT	Seg.	Seeds, Spatial Inf., Anatomy
HAM10000 [36]	10,012	Dermatology	Cls	Seeds
SkinCancer [21]	10,002	Pathology	Cls	-
RibFrac [18]	9,221	CT	Seg.	Seeds, Spatial Inf., Anatomy
VinDr-PCXR [32]	8,939	X-Ray	Cls	Seeds, Visual Desc.
Lung Nodules [2]	8,902	CT	Seg.	Seeds, Spatial Inf., Anatomy
Pulmonary Embolisms [9]	8,614	CT	Seg.	Seeds, Spatial Inf., Anatomy
HyperKvasir-Img [5]	8,204	Colonoscopy	Cls	Visual Desc.
GastroVision [17]	7,930	Endoscopy	BB	Groundings
KiTS19 [13]	7,547	CT	Seg.	Seeds, Spatial Inf., Anatomy
Object-CXR ⁵	7,442	X-Ray	BB	Groundings
EyePACS [37]	6,538	Retinal Image	Cls	-
Diabetic Retinopathy ⁶	3,540	Retinal Image	Cls	-
VinDr-Mammo [31]	3,396	Mammo	Cls	-
MSD Lung [1]	3,296	CT	Seg.	Seeds, Spatial Inf., Anatomy
MSD Colon [1]	2,483	CT	Seg.	Seeds, Spatial Inf., Anatomy
MSD Pancreas [1]	2,117	CT	Seg.	Seeds, Spatial Inf., Anatomy
HyperKvasir-Seg [5]	1,000	Colonoscopy	BB	Seeds, Groundings
MVID	558,052	Mixed	Vis. Instr.	

4.2 Benchmarks

We evaluate the models on four medical benchmarks: VQA-RAD [23], SLAKE [25], PATH-VQA [12], and GMAI-MMBench [7]. VQA-RAD and SLAKE focus on radiology, while PATH-VQA covers pathology images. These datasets include both closed-ended (yes/no) and open-ended questions. Previous evaluations using word recall [24] tend to favor longer answers, while word-based F1 scores were inconsistent [6]. To enhance reliability, we convert open-ended questions into multiple-choice QA (MCQA) format by using an LLM to generate three plausible distractors. This approach preserves question complexity while ensuring a more precise evaluation. We also include GMAI-MMBench for its broad coverage, spanning 284 datasets across 38 medical imaging modalities, 18 clinical tasks, and 18 departments. It consists entirely of MCQA samples.

⁴ <https://dermnet.com/>

⁵ <https://github.com/hlk-1135/object-CXR>

⁶ <https://kaggle.com/competitions/diabetic-retinopathy-detection>

We assess model accuracy by prompting VLMs to begin their answers with either the option of a closed-ended question or the corresponding letter of a multiple-choice response. We then compare the model’s output to the ground truth to compute the accuracy.

Table 2. The results of our comparison of models trained on LLaVA-Med and MVID60k. We report the accuracy on closed and multiple-choice question answering (MCQA).

		VQA-RAD		SLAKE		PATH-VQA		GMAI	AVG
Data	Encoder	Closed MCQA		Closed MCQA		Closed MCQA			
Mistral-7B-Instruct-v0.2									
LLaVA-Med	Clip336	55.38	34.64	62.54	46.20	55.52	36.93	29.00	45.74
MVID60k	Clip336	57.77	32.40	63.70	50.23	59.88	40.09	33.55	48.23
MVID60k	BioClip	61.35	34.70	65.38	51.28	58.66	39.63	37.83	49.83
Biomistral-7B									
MVID60k	Clip336	57.41	46.93	62.74	60.16	57.29	41.36	31.87	51.11
MVID60k	BioClip	59.73	49.16	66.11	62.48	59.19	42.78	36.88	53.76
Qwen2.5-VL-7B-Instruct									
MVID60k	Clip336	64.54	51.96	62.98	69.30	59.58	48.61	37.31	56.33
Llava-Med	BioClip	64.94	55.31	58.41	64.19	55.38	49.31	41.30	55.55
MVID60k	BioClip	68.53	55.87	72.60	69.92	59.73	45.52	42.57	59.25

4.3 Results

Experiment 1 Table 2 presents the results of our first experiment. Training on MVID60k using the same configuration as LLaVA-Med-v1.5 led to improvements in six out of seven metrics, with an average performance gain of 2.49%. Among image encoders, BioCLIP (224×224 resolution) outperformed the general-domain CLIP model (336×336 resolution), highlighting the advantages of a specialized visual encoder on biomedical tasks. Despite being based on the weaker Mistral-7B-Instruct-v0.1, the biomedical LLM Biomistral performed better, particularly when paired with BioCLIP, suggesting a synergistic effect between the two. However, the strongest overall model was Qwen2.5-7B-Instruct, a more recent general-domain LLM. Leveraging these insights, we achieved a **13.51% average performance improvement** over the initial LLaVA-Med-v1.5 checkpoint while using the same amount of data. A performance gap of 3.7% remains when training a model with the same configuration on the LLaVA-Med dataset. Additionally, our MCQA reformulations appear to enhance expressiveness, particularly in PATH-VQA, where the MCQA version produced a wider range of results compared to the closed-question format.

Experiment 2 We present the results of our second experiment in Table 3. By training on the entire MVID dataset, we increase the **average performance improvement to LaVA-Med-v1.5 to 20.29%**. The model trained on MVID outperforms most general and medical VLMs included in our comparison and surpasses GPT-4o on VQA-RAD and the closed-question subset of SLAKE. HuatuoGPT-Vision-7B [6] performs better on non-radiological tasks, however, it was trained on a large dataset generated by GPT-4 Vision, an already strong VLM. In contrast, our approach is based on a text-only, open-weight LLM, demonstrating competitive performance despite more constrained and privacy preserving resources.

Table 3. The results of our comparison with between our best model and different existing VLMs.

Model	VQA-RAD		SLAKE		PATH-VQA		GMAI		AVG
	Closed	MCQA	Closed	MCQA	Closed	MCQA	Closed	MCQA	
General VLMs									
LLaVA-v1.6-7B [26]	52.99	35.75	54.81	48.22	51.61	46.03	35.44		48.23
LLaVA-v1.6-13B [26]	61.35	38.55	54.09	55.04	54.91	47.21	36.63		51.86
LLaVA-v1.6-34B [26]	63.35	55.31	56.49	66.20	64.40	49.24	38.19		59.16
GPT4o	66.53	64.25	71.63	74.57	75.97	52.49	53.96		67.57
Medical VLMs									
Med-Flamingo [28]	54.18	25.70	52.88	21.86	56.81	23.75	11.64		39.20
Llava-Med v1.5[24]	55.38	34.64	62.54	46.20	55.52	36.93	29.00		45.74
HuatuoGPT-Vision-7B [6]	70.59	64.80	75.42	75.66	59.29	54.55	50.55		67.16
MVID-Qwen2.5-7B	72.06	65.36	73.25	73.49	61.54	50.47	44.06		66.03

5 Discussion

Our results demonstrate the feasibility of training a competitive medical VLM by leveraging labeled medical imaging data and an open-weight LLM. This finding is particularly relevant in clinical research environments, where data privacy restrictions usually prohibit the use of commercial LLMs. In such settings, researchers frequently have access to labeled imaging datasets, making our approach a viable alternative to proprietary solutions.

Furthermore, our study highlights that despite not utilizing a stronger VLM for data generation, our model’s performance remains comparable to those that integrate GPT-based VLMs. This suggests that an open-weight LLM, when combined with labeled imaging data, can effectively support medical VLM training, denying the dependence on proprietary models.

One notable observation is the performance bias towards radiological tasks, which may stem from their overrepresentation in MVID. In contrast, for other

imaging modalities, our model often relied on classification label-based information due to the limited availability of more detailed annotations. This suggests that expanding labeled datasets across diverse medical imaging domains could further enhance the model generalizability.

In future work, extending our approach with an open-weight VLM could provide additional benefits. By incorporating a more advanced vision model, we anticipate improvements in both multimodal understanding and task performance, particularly in underrepresented imaging modalities. Additionally, investigating strategies to balance dataset composition and annotation quality across modalities could mitigate performance biases and enhance model robustness in broader clinical applications.

6 Conclusion

In conclusion, our results demonstrate that it is possible to train competitive medical vision-language models using only labeled imaging datasets, without relying on proprietary LLMs. By leveraging MVID, a dataset generated with an open LLM, we show that medical VLMs achieve strong performance on medical VQA benchmarks and perform competitively with state-of-the-art VLMs.

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